Sparse-Concept Calibration of Knowledge Tracing Models for Threshold-Based Educational Decisions

Khanh-Trinh Nguyen[1], Tuan Dao Minh[1], Duong Nguyen Tien[1], Chi Thanh Nguyen[2], and Van-Hau Nguyen[1,*]
1 Hung Yen University of Technology and Education, Hung Yen, Vietnam
2 Academy of Military Science and Technology, Ha Noi, Vietnam
Email: trinhnk@utehy.edu.vn (K.-T.N.); tuanymc@utehy.edu.vn (T.D.M.); duongnt@utehy.edu.vn (D.N.T.); thanhnc@ioit.ai.vn (C.T.N.); haunv@utehy.edu.vn (V.-H.N.)
*Corresponding author

Abstract—*Knowledge Tracing (KT) systems that skip, remediate, or advance practice typically consume a predicted probability rather than an area-under-the-curve (AUC) score, so population ranking can look acceptable while probabilities are poorly calibrated on rarely practiced knowledge components (KCs). This paper reports a diagnostic evaluation—not a new KT architecture—of train-only KC-frequency strata on three public logs (ASSISTments 2012, Junyi Academy, and XES3G5M) with IRT, DKT, and SimpleKT. Lower KC training frequency does not universally degrade discrimination, but calibration can become less reliable in some sparse-concept regimes. On ASSISTments 2012, SimpleKT expected calibration error (ECE) rises from 0.114 on dense KCs to 0.228 on sparse KCs (Limited occupancy, N≈415); Junyi has no learner-based sparse stratum, and XES3G5M SimpleKT ECE is essentially flat. A locked global threshold at τ=0.7 raises the SimpleKT false-advance rate (incorrect-response rate among advance decisions) from 0.196 (dense) to 0.268 (sparse) on one ASSISTments fold; the sparse–dense gap stays positive on all five training seeds (mean 0.047; four unique student partitions). This is a simulated decision gate, not a classroom intervention. Probability-threshold decisions should be validated by KC-frequency stratum rather than on population AUC or ECE alone.*

Keywords—*knowledge tracing, calibration, sparse concepts, learning analytics, educational decision support, mastery threshold*

I. Introduction

Adaptive practice platforms and intelligent tutoring systems use Knowledge Tracing (KT) to estimate a learner's command of a skill and to decide whether to skip, remediate, or advance practice [1]–[3]. Those systems rarely consume a population area-under-the-curve (AUC) statistic. They consume a predicted probability p, which is then compared with a threshold τ. If $p \geq \tau$, the platform typically withholds additional practice on that skill. The operational chain is therefore population AUC → predicted probability → calibration → threshold-based educational decisions, with particular diagnostic risk on sparsely trained knowledge components (KCs).

KT evaluations typically report population AUC and accuracy [4], [5]. Those metrics rank models, but they can conceal miscalibration on low-frequency KCs—skills with little training-fold evidence—because most test events lie on dense, frequently practiced concepts. A model can discriminate well in aggregate and still assign overconfident probabilities on the sparse tail. When a fixed threshold is applied to p, that overconfidence appears as an advance decision followed by an incorrect next response. We term this the false-advance rate (FAR). Section III defines FAR as $P(y=0 \mid p \geq \tau)$; y denotes observed next-response correctness, not latent mastery truth.

This paper treats that mismatch as an evaluation problem for educational technology, not as a reason to propose another KT architecture. The study is organized around three research questions. RQ1: Does lower KC training frequency systematically degrade predictive discrimination? RQ2: How does calibration vary across KC-frequency strata and datasets? RQ3: When a fixed probability threshold is applied, does decision-error behavior differ between sparse and dense KCs?

The empirical answers are bounded. Lower KC training frequency does not universally degrade discrimination: on XES3G5M, sparse AUC is higher than dense AUC for DKT and SimpleKT. Calibration can, however, become less reliable in some sparse-concept regimes. On ASSISTments 2012, SimpleKT expected calibration error (ECE) increases from dense to sparse KCs, and a locked gate at τ=0.7 yields a higher FAR on sparse than on dense advances. The same ECE gradient is absent on Junyi, where the learner-based sparse stratum is empty, and is essentially absent for SimpleKT on XES3G5M. We therefore do not claim that sparse KCs always fail, and we do not claim a causal effect of training frequency on calibration.

Contributions are conservative: (i) a train-only KC-frequency protocol with an explicit strict cold-start group, so the definition of "sparse" cannot leak test-fold counts; (ii) per-stratum calibration (ECE and Brier decomposition) on three public datasets, with occupancy flags (Reliable / Limited / Insufficient); (iii) a locked-threshold simulation of FAR and miss rates, with a five-seed check of the sparse–dense FAR gap on ASSISTments 2012. We do not propose a new KT architecture, a new calibration algorithm, or a new auditing theory, and we do not report a classroom intervention. A graph-KT (GKT) model and a CL4KT-style adapter appear only as an exploratory single-fold diagnostic on ASSISTments.

II. Literature Review

A. Knowledge Tracing and benchmark models

Knowledge Tracing (KT) models a learner's evolving proficiency from historical interactions in order to predict the next response [3]. Classical approaches include Bayesian Knowledge Tracing (BKT), which tracks a latent mastery state for each skill [1], and item-response theory (IRT), of which the one-parameter Rasch model is the canonical form used in this study [6]. Deep Knowledge Tracing (DKT) replaced hand-specified mastery dynamics with a recurrent sequence model and established the modern next-response prediction task [2]. Attention-based successors, including context-aware attentive KT [7] and SimpleKT [4], retain that task while substituting self-attention for recurrence. Benchmarking libraries such as pyKT have standardized preprocessing and comparison, and have shown that evaluation choices can distort reported discrimination [5]. Across this line of work, the primary reported statistic remains population AUC.

B. Graph and self-supervised KT

Graph-based KT (GKT) represents knowledge components as nodes of a graph and updates proficiency with a graph neural network [8]. Contrastive learning for KT (CL4KT) trains representations from augmented views of learning histories [9]. These models are related architectures, not the contribution of the present paper. They are included later only as an exploratory single-fold diagnostic on ASSISTments 2012, not as a proposed method and not as a state-of-the-art comparison.

C. Sparse-data and cold-start problems

The term "sparse" is used in several incompatible senses in the KT literature. This paper distinguishes four. (1) Sparse attention: sparseKT applies k-sparse attention so that only a subset of historical interactions receives nonzero weight [16]. (2) Sparse KC frequency: a knowledge component with few training-fold observations. (3) New-student cold start: prediction for learners who have little or no interaction history at test time [17]. (4) Concept-level cold start: a knowledge component with zero training-fold frequency, regardless of whether the learner is new. Educational interaction logs additionally exhibit a long tail in which a small set of dense KCs dominates event volume [20].

These four problems are not interchangeable. sparseKT-style sparse attention is not equivalent to low-frequency KCs: attention can be sparsified on a frequently practiced skill. New-student cold start is not the same as concept-level zero-frequency cold start: a previously observed learner may still encounter a concept that never appeared in the training fold. The protocol in Section III operationalizes (2) and (4) through train-only KC-frequency strata with an explicit strict cold-start group (f=0).

D. Calibration and educational decision support

When predicted probabilities are consumed as educational decisions, ranking is not a sufficient evaluation [10]. Calibration asks whether the predicted probability p matches the empirical frequency of a correct next response. Expected calibration error (ECE) is the usual bin-wise summary of that mismatch [12]; modern neural networks can remain poorly calibrated even when they discriminate well [11]. The Brier score [13] and its reliability, resolution, and uncertainty decomposition [14] separate miscalibration from task difficulty, and reliability diagrams display the same probability-level comparison. Complementary post-hoc work has begun to correct per-item bias after a frozen KT backbone [15]. Probability-level evaluation of this kind is still typically reported on the pooled population, not on train-only frequency strata.

Existing KT benchmarks mainly emphasize aggregate discrimination. Sparse frequency, calibration, sample support, and threshold-based decision error are rarely examined together. This study keeps standard models and reports those four quantities jointly, so that a population AUC result can be read as an educational-technology decision rather than only as a leaderboard rank.

III. Materials and Methods

A. Datasets

We use three de-identified public logs after a common interaction schema: ASSISTments 2012 [18], [5], Junyi Academy [19], and XES3G5M [20]. Table 1 reports post-processing counts: processed learners, processed KCs, processed interactions, and learner-based test events. Exact processed interaction totals are 2,657,490 (ASSISTments 2012), 16,215,567 (Junyi Academy), and 7,953,709 (XES3G5M). Preprocessing drops rows with missing learner, KC, or label identifiers, binarizes correctness, parses timestamps, and keeps learners with at least two interactions. Those filters—not the raw public dumps—are what Table 1 counts.

Provenance. ASSISTments 2012 is the 2012–2013 public release with affect [18], documented for KT use in pyKT [5]; each retained row already carries one skill_id. Junyi Academy is the Kaggle Online Learning Activity Dataset [19] (not Junyi2015); the KC field is ucid. XES3G5M [20] is loaded from the authors' kc_level split: train_valid_sequences.csv and test.csv are concatenated and flattened to one row per sequence position.

XES3G5M counts. The original dataset paper reports 18,066 students, 7,652 questions, 865 KCs, and 5,549,635 question-level interactions [20]. Table 1 instead lists 866 KCs and 7.95M interactions because flattening kc_level sequences (i) expands multi-KC questions into one row per listed concept and (ii) retains sequence-padding tokens coded skill_id=−1 (with matching question_id=−1). Unique kc_id including that padding token is 866; excluding it is 865. Unique item_id including −1 is 7,653. Subsequent preprocessing dropped 0 rows. We do not retabulate Table 1 to the question-level official totals.

TABLE 1. POST-PROCESSING COHORT STATISTICS (LEARNER-BASED SPLIT). DISPLAYED INTERACTION TOTALS MATCH THE PROCESSED COUNTS 2.66M, 16.2M, AND 7.95M.

IV. DATASET	V. LEARNERS	VI. KCS	VII. INTERACTIONS	V[illegible]
IX. ASSISTMENTS 2012	X. 27,806	XI. 265	XII. 2.66M	XII[illegible]
XIV. JUNYI ACADEMY	XV. 71,014	XVI. 1,326	XVII. 16.2M	XVII[illegible]
XIX. XES3G5M	XX. 18,066	XXI. 866	XXII. 7.95M	XXII[illegible]

B. Splits and seeds

The learner-based split is primary. Users are shuffled with the fold seed; 20% of learners are held out as test, 10% as validation, and the remainder as training. Within a fold, train, validation, and test learners are disjoint. The temporal split is complementary: all interactions are sorted by timestamp; the earliest 70% are training, the next 10% validation, and the latest 20% test. Gate numbers below are not taken from the temporal split. KC-frequency buckets are constructed from the training file only.

Deep models are trained in five runs at seeds 42, 2024, 2025, 2026, and 2027 (fold_0 through fold_4). Seeds 2025 and 2026 share one student partition (fold_2 = fold_3) on all three datasets. These are five training runs and four unique student partitions, not five independent folds. Tables that report mean±sd therefore summarize four unique partitions: the two initializations on the duplicated split are averaged first. The gate-robustness table still lists all five training seeds.

C. Model settings

Main tables use local IRT, DKT, and SimpleKT implementations, not a pinned pyKT commit (training snapshot commit: NOT RECOVERED). The local SimpleKT class is a two-layer Transformer encoder and is not byte-identical to the official SimpleKT checkpoint [4]. Recovered hyperparameters follow; missing fields are marked NOT RECOVERED and are not imputed.

RECOVERED TRAINING SETTINGS FOR THE MAIN BASELINES. NOT RECOVERED CELLS WERE NOT IMPUTED. THIS LISTING IS NOT A RESULTS TABLE.

XXIV. SETTING	XXV. IRT 1PL	XXVI. DKT	XXVII. SIMPLEKT
XXVIII. IMPLEMENTATION	XXIX. LOCAL IRT	XXX. LOCAL DKT	XXXI. LOCAL SIMPLEKT
XXXII. VERSION/COMMIT	XXXIII. NOT RECOVERED	XXXIV. NOT RECOVERED	XXXV. NOT RECOVERED
XXXVI. MAJOR HYPERPARAMETERS	XXXVII. 1PL: SIGMOID(Θ−B); L2 0.01	XXXVIII. LSTM EMBED 64, HIDDEN 128	XXXIX. TRANSFORMER 2 LAYERS, 4 HEADS, EMBED 64
XL. OPTIMIZER	XLI. SGD	XLII. ADAM	XLIII. ADAM
XLIV. LEARNING RATE	XLV. 0.01	XLVI. 1E−3	XLVII. 1E−3
XLVIII. EARLY STOPPING	XLIX. NONE (10 EPOCHS)	L. NONE (50 EPOCHS)	LI. NONE (50 EPOCHS)
LII. BATCH SIZE	LIII. 512	LIV. 64	LV. 64
LVI. MAX. SEQUENCE LENGTH	LVII. N/A	LVIII. 200	LIX. 200
LX. SELECTION METRIC	LXI. LAST EPOCH	LXII. VALIDATION AUC	LXIII. VALIDATION AUC

On ASSISTments fold 0 only we also score a train-only GKT graph [8] (pyKT GKT; Adam, learning rate 1e−3, batch size 16, maximum length 100, hidden and embedding size 32, dropout 0.5, 20 epochs, patience 4, selection by validation AUC) and a CL4KT protocol adapter [9] (contrastive views on training sequences; not an official CL4KT checkpoint; Adam, learning rate 1e−3, batch size 64, maximum length 100, hidden size 64, 4 heads, 2 layers, dropout 0.2, 20 epochs, patience 6, selection by validation AUC). IRT under learner-based splits has no ability parameter for unseen students, so its AUC is 0.50 by construction; we report it as a base-rate reference, not as a ranking competitor.

D. Train-only frequency strata

For each fold, KC frequency f_train is counted on the training file only. Buckets: strict cold-start (f=0), very sparse (0<f<20), sparse (20≤f<100), medium (100≤f<500), dense (f≥500). Dense KCs dominate event volume while a non-empty sparse-like tail exists on ASSISTments and XES3G5M.

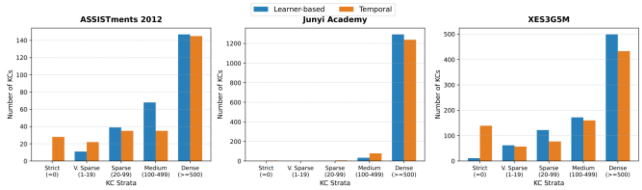

Fig. 1. Train-only KC-frequency strata. Dense concepts dominate interactions; sparse and cold-start KCs remain the diagnostic tail.

E. Reliability flags

Occupancy flags follow test-event count N: Reliable (R) N≥1000; Limited (L) 100≤N<1000; Insufficient (I) N<100. We call R, L, and I descriptive sample-support flags. They are not inferential guarantees, confidence intervals, or hypothesis tests. Success claims require Limited or Reliable occupancy. Insufficient cells are descriptive only. Very-sparse ASSISTments cells are Insufficient and descriptive only.

F. Calibration

Let y∈{0,1} be next-response correctness and p∈[0,1] the predicted probability. With M=15 equal-width bins, ECE = Σ_m (n_m / N) |acc_m − conf_m|. The Brier score is (1/N) Σ_i (p_i − y_i)². We use the binned decomposition Brier = UNC − RES + REL [14], with UNC = ȳ(1−ȳ), REL = (1/N) Σ_m n_m (conf_m − acc_m)², and RES = (1/N) Σ_m n_m (acc_m − ȳ)². Because probabilities are binned, the three components are an empirical approximation and need not sum exactly to the directly computed Brier score. Lower ECE is better calibration; it is not a substitute for AUC.

G. Difficulty coupling

For each KC c we define a training-only difficulty proxy difficulty(c) = 1 − mean_train_correctness(c). The mean is taken on the training file only. This is an observational proxy for how often c is answered incorrectly in training; it is not a latent IRT difficulty parameter. The reported Spearman association is ρ(log(1+f_train), difficulty), computed on training-fold KC summaries. The association is descriptive and is not a causal effect of frequency on difficulty or on calibration.

H. Simulated decision gate

For a locked global threshold τ, the system advances (skips remediation) if p≥τ and triggers remediation otherwise. Sparse events never receive their own tuned τ. The display threshold is τ=0.7; the grid {0.5, 0.6, 0.7, 0.8} is recorded in the artifact and is not a search over sparse error.

Let A={p≥τ}. The false-advance rate (FAR) is P(y=0 | A): among advances, how often the next response was incorrect. y is observed next-response correctness, not latent mastery. Miss is P(A | y=0): among incorrect answers, how often the system still skipped help. Result tables label FAR as false mastery (FM); the definition is the same. If the model were calibrated on the advance set, FAR would match E[1−p | A]. We report ΔFM = FM_sparse − FM_dense. Positive ΔFM means sparse advances are dirtier than dense advances. This is a simulated decision error, not an instructional RCT.

LXIV. RESULT AND DISCUSSION

A. Aggregate Ranking Is Not the Sparse Story

Table 2 shows learner-based AUC. DKT is slightly above SimpleKT on all three datasets. IRT AUC is 0.50 for the reason given above. These numbers would look like a routine KT bake-off. They do not tell a designer whether p is trustworthy on the tail.

TABLE 2. OVERALL LEARNER-BASED AUC (MEAN±SD OVER FOUR UNIQUE PARTITIONS)

Dataset	Model	AUC	ACC
ASSISTments 2012	IRT	0.5000	0.6973±0.0004
ASSISTments 2012	DKT	0.6979±0.0014	0.7182±0.0014
ASSISTments 2012	SimpleKT	0.6837±0.0025	0.6996±0.0032
Junyi Academy	IRT	0.5000	0.7053±0.0018
Junyi Academy	DKT	0.7320±0.0009	0.7343±0.0015
Junyi Academy	SimpleKT	0.7231±0.0030	0.7274±0.0015
XES3G5M	IRT	0.5000	0.7961±0.0031
XES3G5M	DKT	0.8171±0.0022	0.8327±0.0032
XES3G5M	SimpleKT	0.7557±0.0013	0.8067±0.0037

On XES3G5M, sparse AUC is higher than dense AUC (DKT 0.857 vs. 0.817; SimpleKT 0.847 vs. 0.755; Reliable occupancy). Lower frequency is therefore not a universal ranking failure. Junyi's learner-based sparse bucket is empty under the pre-registered cuts—a protocol outcome, not a missing cell to impute.

B. Calibration Can Move When Ranking Does Not

Table 3 is the calibration punchline. On ASSISTments 2012, SimpleKT ECE increases from 0.1136±0.0066 (dense, Reliable, N=523,971) to 0.1541 (medium) to 0.2280±0.0197 (sparse, Limited, N=415). DKT shows the same direction (dense ECE 0.060 vs. sparse 0.233). IRT dense ECE is tiny (0.003) but resolution is zero: a constant-like predictor can look "calibrated" while ranking nothing.

On Junyi, only dense and medium strata exist; SimpleKT ECE rises modestly from 0.079 to 0.107. On XES3G5M, SimpleKT ECE is essentially flat (0.114→0.111→0.125 dense/medium/sparse) despite Reliable sparse occupancy (N=2,010). ECE-flat is not a license to skip occupancy reporting, and—as the gate results show—it is not the same as a flat miss rate.

TABLE 3. SIMPLEKT EVENT-LEVEL ECE BY TRAIN-ONLY FREQUENCY STRATUM (FOUR UNIQUE PARTITIONS). FLAGS: R RELIABLE, L LIMITED. JUNYI SPARSE IS EMPTY.

Dataset	Stratum	N	Flag	ECE
ASSISTments 2012	dense	523,971	R	0.1136±0.0066
ASSISTments 2012	medium	5,963	R	0.1541±0.0051
ASSISTments 2012	sparse	415	L	0.2280±0.0197
Junyi Academy	dense	3,232,614	R	0.0792±0.0051
Junyi Academy	medium	3,836	R	0.1073±0.0156
XES3G5M	dense	1,268,696	R	0.1145±0.0011
XES3G5M	medium	12,980	R	0.1114±0.0076
XES3G5M	sparse	2,010	R	0.1248±0.0085

C. A Global Gate Can Turn the ASSISTments Gradient into a Decision Error

Table 4 applies τ=0.7 on ASSISTments fold 0 (seed 42; sparse N=444, Limited). SimpleKT false mastery among advances is 0.196 on dense KCs and 0.268 on sparse KCs (ΔFM=+0.072). The calibrated expectation E[FM] on sparse advances is only 0.050, so the excess error is large. DKT shows a similar FM gap. Train-only GKT shrinks ΔFM to +0.015 (FM 0.205 → 0.220). The CL4KT adapter is intermediate (0.185→0.240). We do not read this as "use GKT in production"; we read it as: the ASSISTments calibration gradient is not a property of the dataset alone.

TABLE 4. SIMULATED GATE AT T=0.7, ASSISTMENTS 2012 FOLD 0. ADVANCE IF P≥T. NOT A CLASSROOM TRIAL.

Model	Stratum	N	FM	E[FM]	Miss
SimpleKT	dense	528,018	0.196	0.113	0.352
SimpleKT	sparse	444	0.268	0.050	0.320
DKT	dense	528,018	0.200	0.147	0.398
DKT	sparse	444	0.296	0.057	0.365
GKT (train-only)	dense	528,018	0.205	0.163	0.455
GKT (train-only)	sparse	444	0.220	0.157	0.234
CL4KT (adapter)	dense	528,018	0.185	0.175	0.359
CL4KT (adapter)	sparse	444	0.240	0.116	0.244

Table 5 checks whether SimpleKT ΔFM is a one-fold accident. It is positive on all five training seeds (mean 0.047, sd 0.033). A KC-clustered bootstrap on seed 42 yields a 95% interval [0.006, 0.138]—excluding 0, but wide, as Limited occupancy requires. DKT ΔFM is positive on only three of five seeds and is not treated as a five-run finding. On XES3G5M, SimpleKT ΔFM is negative on all five seeds, but ΔMiss is positive on all five (mean +0.112): among actual incorrect answers, the system still advances more often on sparse than on dense KCs. A flat ECE therefore does not imply a flat miss rate.

Coincidence of digits: SimpleKT dense E[FM] at τ=0.7 on seed 42 is 0.113, close to the four-partition dense ECE 0.114. They are different quantities.

TABLE 5. GATE ROBUSTNESS AT T=0.7 ON ASSISTMENTS 2012 (FIVE TRAINING SEEDS; FOUR UNIQUE PARTITIONS). GKT/CL4KT REMAIN SEED 42 ONLY.

Model	Mean ΔFM	SD	Seeds >0	Mean sparse N
SimpleKT	0.047	0.033	5/5	413
DKT	0.033	0.048	3/5	413

D. What a Designer Should Take Away

If a learning platform uses KT probabilities only to sort students, Table 2 may suffice. If it uses p to skip practice or declare mastery, Table 2 is the wrong dashboard. The ASSISTments SimpleKT cell is the cautionary case: competitive AUC, Limited-but-estimable sparse occupancy (about 19% of KCs fall below the sparse threshold on fold 0), a monotonic ECE rise, and a gate that advances more dirty sparse attempts. The XES3G5M cell is the opposite caution: plenty of sparse mass and Reliable occupancy, yet SimpleKT ECE does not degrade—while miss rates still can. Junyi shows that the protocol can refuse a sparse claim when the bucket is empty.

Three observable conditions track when sparse-calibration claims are informative rather than obligatory (Table 6): (1) a non-empty low-frequency tail under the split actually used; (2) enough sparse test events for at least a Limited-flag ECE; (3) frequency–difficulty coupling that is not inverted. ASSISTments meets all three. That is a diagnostic pre-condition, not a claim that training frequency causes miscalibration.

TABLE 6. WHEN A SPARSE-CALIBRATION CLAIM IS ESTIMABLE (FINDINGS, NOT CAUSES). SPARSE MASS: SHARE OF KCS WITH F_TRAIN<100.

Condition	ASSISTments	Junyi	XES3G5M
Sparse mass	18.9% of KCs	0%	22.5% of KCs
Sparse N	415 (L)	empty	2,010 (R)
Difficulty coupling	ρ=−0.227 (sparse harder)	ρ=−0.416	ρ=+0.087 (weak, inverted)
SimpleKT ECE	0.114→0.228	dense→medium only	flat 0.114→ 0.125

E. *What This Paper Does Not Show*

The gate is not a classroom policy and not an A/B test of remediation. GKT and the CL4KT adapter are single-fold, ASSISTments-only instantiations. Temporal results are a single corrected cutoff (seed 42), not a multi-cutoff variance estimate. IRT is not a fair ranking baseline on unseen learners. We did not tune τ on sparse events, and we do not recommend doing so in deployment without a separate validation design.

F. *Implications for Information and Education Technology*

IJIET's audience includes operators of educational platforms, not only KT researchers. For that audience, the operational recommendation is narrow: (i) log train-only KC frequency with every prediction; (ii) report ECE and occupancy on dense vs. sparse slices before enabling a global mastery threshold; (iii) if a threshold must be global, simulate FM and Miss on the sparse slice at the same τ used in the product; (iv) do not treat a population AUC win, or a graph/contrastive architecture, as automatically safer on the tail.

LXV. CONCLUSION

KT models that look adequate on AUC can still be poorly calibrated on rarely practiced skills, and a global mastery threshold can then produce a higher false-mastery rate on those skills. That pattern appears for SimpleKT on ASSISTments 2012 (ECE 0.114→0.228, Limited N=415; seed-42 FM 0.196→0.268; △FM>0 on 5/5 seeds). It does not appear as a universal law: Junyi has no learner-based sparse stratum, and XES3G5M SimpleKT ECE is flat. Sparse-concept diagnostics are therefore conditionally important. The simulation is a decision-error check for educational-technology systems, not a claim of a new KT model and not a classroom intervention.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

K.-T.N. led the diagnostic design, experiments, and manuscript. T.D.M. and D.N.T. contributed data processing and baseline runs. C.T.N. contributed methodological review. V.-H.N. supervised the study and revised the manuscript. All authors approved the final version.

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Anonymized code and prediction exports are available for peer review; a public repository will be released upon acceptance. This study uses de-identified public learner logs (ASSISTments 2012, Junyi Academy, XES3G5M) and is not a classroom trial.

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